

Revision of SPM 1984 Wave Hindcast Model to Avoid Inconsistencies in Engineering Applications

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ABSTRACT

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Inconsistencies in the Shore Protection Manual (SPM) formulations for the sea state generated by a constant wind blowing over a fetch are identified. An alternative single set of formulations is given which asymptotically matches the SPM formulations and which is generally easier to apply. Modified expressions for the significant wave height, period of peak spectral energy and limiting duration are given. Graphical comparison is made with the original formulations.

1. INTRODUCTION

Simplified wave hindcast models are widely used to estimate the wave climate at locations where no wave data are available. A commonly used model is that given in the Shore Protection Manual. This deals with the waves generated by a uniform wind blowing over a known fetch for a given duration. The 1984 edition of the manual revised these equations to account for new data which were available.

The manual presents different equations for three cases, deep water developing seas, deep water fully developed seas and shallow water. Unfortunately the equations presented do not match at the transition points between each of the cases. Further in very shallow water the relationship between critical duration and fetch is also questionable.

The purpose of this paper is to identify how the equations do not match and to present an alternative single set of equations to replace the three cases given in the SPM (1984). These equations asymptotically match the SPM (1984) equations in the case of very deep or very shallow water and in the case of very short or very long fetch. The predicted values also remain between the values predicted in the various cases at intermediate values of fetch and depth. A

revised relationship between the critical duration and fetch is also presented, based on physical considerations.

It is hoped that this will make the equations easier to apply and will remove the need for decisions about which set of equations should be applied. The use of a single set of equations also avoids step changes of predicted properties with small changes in the input parameters. Such step changes can cause problems in engineering situations.

The background to the formulations given in the SPM (1984) is not repeated here. In this note the same notation is used as in the SPM (1984) but in general only the non-dimensional equations are presented and additional variables are introduced to represent non-dimensionalised parameters.

2. NON-DIMENSIONALISATION

Distance, height and depth are non-dimensionalised by multiplying by a factor:

$$\frac{g}{U_A^2}$$

Time is non-dimensionalised by multiplying by a factor:

$$\frac{g}{U_A}$$

It is assumed that consistent unit systems are used for all quantities. U_A is the wind stress factor (adjusted windspeed) calculated from the windspeed over open water, adjusted for air/water temperature difference, using the formulae given in the SPM (1984):

$$U = R_T U(10) \quad (2.1)$$

$$U_A = 0.71 (U)^{1.23} \quad (2.2)$$

$U(10)$ is the mean hourly windspeed 10 m above mean sea level and R_T an air-sea temperature difference adjustment factor (default value 1.1).

g is the acceleration due to gravity.

A variable that has been non-dimensionalised using the conventions above is denoted by the same symbol as for the original variable with an overhead $\sim$. For example:

$$\tilde{T}_m = \frac{g T_m}{U_A}$$

where T_m is the period of peak spectral energy measured in seconds.

3. REVIEW OF SPM FORMULATIONS

3.1 Statement of the formulations

In this section the SPM formulations for both deep and shallow water are outlined without comment for the benefit of the reader.

The limited sea state equations for deep water are given in the SPM (1984) on page 3–48 as follows:

$$\tilde{H}_s = 0.0016\tilde{F}^{1/2} \quad (3.1)$$

$$\tilde{T}_m = 0.286\tilde{F}^{1/3} \quad (3.2)$$

$$\tilde{t}_{lim} = 68.8\tilde{F}^{2/3} \quad (3.3)$$

where: H_s is the significant wave height in metres,

F is the fetch in metres,

T_m is the period of peak spectral energy in seconds,

t_{lim} is the limiting duration in seconds.

If the duration of the storm is less than t_{lim} the sea state is duration limited and values of H_s and T_m must be based on a modified fetch length calculated from eqn. 3.3:

$$\tilde{F}' = \left(\frac{\tilde{t}}{68.8} \right)^{3/2} \quad (3.4)$$

However, if both $\tilde{t}$ and $\tilde{t}_{lim}$ are greater than $\tilde{t}_{full}$ then the sea state is fully developed. $\tilde{t}_{full}$ is given by:

$$\tilde{t}_{full} = 71,500 \quad (3.5)$$

In this case $\tilde{H}_s$ and $\tilde{T}_m$ must be calculated according to

$$\tilde{H}_s = 0.243 \quad (3.6)$$

$$\tilde{T}_m = 8.134 \quad (3.7)$$

The fetch-limited sea state equations for use in shallow water are given in the SPM (1984) on page 3–55 as follows:

$$\tilde{H}_s = 0.283 \tanh (0.53\tilde{d}^{3/4}) \tanh \left\{ \frac{0.00565\tilde{F}^{1/2}}{\tanh (0.53\tilde{d}^{3/4})} \right\} \quad (3.8)$$

$$\tilde{T}_m = 7.54 \tanh (0.833\tilde{d}^{3/8}) \tanh \left\{ \frac{0.0379\tilde{F}^{1/3}}{\tanh 0.833\tilde{d}^{3/8}} \right\} \quad (3.9)$$

where d is the (uniform) water depth in metres.

$$\tilde{t}_{lim} = 537\tilde{T}_m^{7/3} \quad (3.10)$$

If the duration of the storm is less than t_{lim} the sea state is duration limited and values of H_s and T_m must be based on a modified fetch calculated from eqns. 3.9 and 3.10.

3.2 Appraisal of the SPM formulations for H_s and T_m

In this section the deep and shallow water SPM (1984) formulations are critically examined. It is first assumed that all sea states are fetch limited, as the relationship between the limiting duration, and fetch is examined hereafter.

Figures 1 and 2 show the variation with fetch of significant wave height and period of peak spectral energy calculated according to the deep water, SPM (1984) equations for a wind of 15 m/s ($U_A = 22.3$ m/s). Note that values of fetch shown are unrealistically large. However the performance of the equations at these fetches is still relevant for higher values of the wind speed because of the non-dimensionalisations used in the formulations.

The transition to the fully developed equations is not occurring in Figs. 1 and 2 at the fetch given by eqns. 3.4 and 3.5 but at an improved fetch obtained by replacing 3.5 by:

$$\tilde{t}_{\text{full}} = 55,800 \quad (3.11)$$

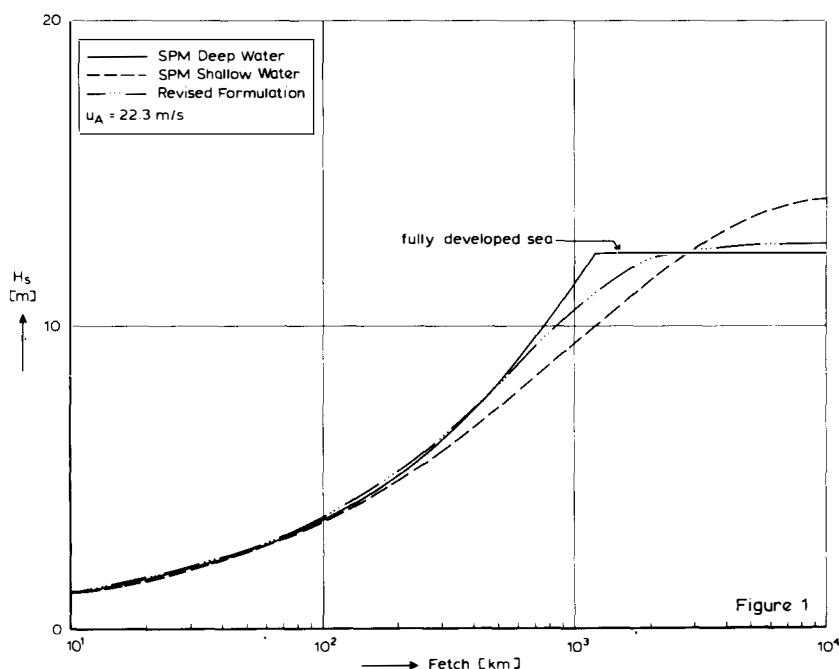


Fig. 1. Variation of predicted significant wave height with fetch for water depth of 10,000 m and unlimited duration.

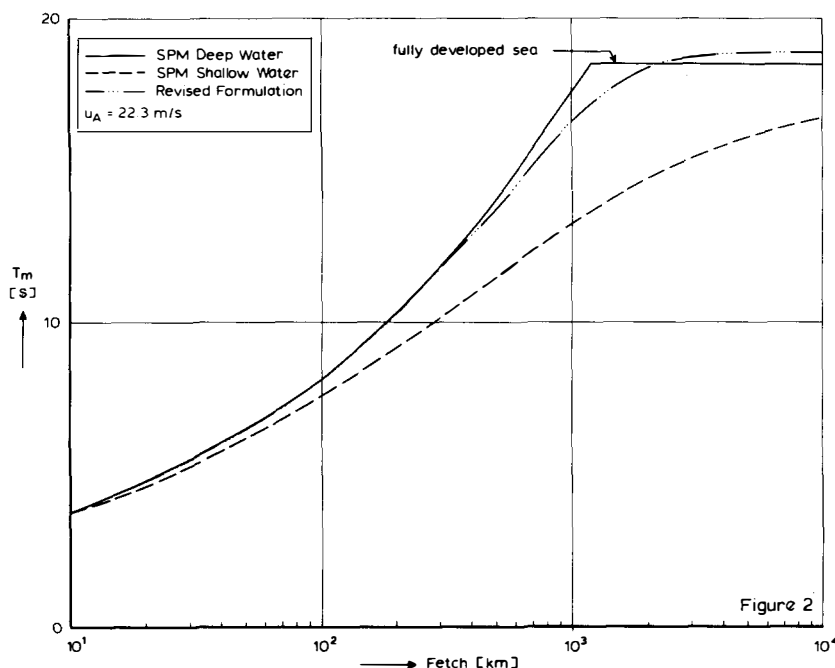


Fig. 2. Variation of predicted period of peak spectral energy with fetch for water depth of 10,000 m and unlimited duration.

giving $F = 1,171$ km according to eqn. 3.4.

This is consistent with the plots given in the nomogram on page 3.49 of the SPM (1984) and it is assumed that a misprint has taken place.

Figure 1 also shows the variation of the predicted significant wave height with fetch for the shallow water equations (3.8) using a value for water depth of 10,000 m so that the water can be considered deep. This shows that for low fetch lengths there is good agreement between the two formulations but the shallow water equation gives results which are up to 20% lower at intermediate fetch lengths and 16% higher at long fetch lengths. Similar discrepancies are found in the expressions for T_m (see Fig. 2). This means that in situations where the effect of water depth is being examined it is inevitable that there will be a step change in the results in the transition between the use of the shallow and the deep water formulations.

3.3 Appraisal of the SPM formulations for limiting duration

The sea state can only increase to the predicted level if the wind continues to blow for the length of time that it takes for the waves to travel the fetch. If the wind dies down when the waves have only travelled a distance, F' (less than the fetch, F) then the sea state is duration limited and F' should be used in the formulations in place of F .

The hindcast model computes irregular waves, characterized by H_s and T_m . Therefore, the model implicitly assumes that all the energy of a sea is concen-

trated around the period of peak spectral energy, T_m . In order to describe the storm speed, a good choice would be the group velocity associated with T_m .

Then the theoretical limiting duration, t_{lim} , associated with a fetch F can be approximated by:

$$t_{lim}(F) = \int_0^F \frac{1}{V_g(F^*)} dF^* \quad (3.12)$$

where $V_g(F)$ is given by:

$$V_g(F) = \left(\frac{1}{2} + \frac{kh}{\sinh 2kh} \right) \frac{\omega}{k} \quad (3.13)$$

where $k \tanh kh = \omega^2/g$

h = water depth

$\omega = 2\pi/T_m(F)$

$\frac{T_m(F)}{g \frac{U_A}{U_A}} = \xi(\tilde{F})$

and ξ is a function which depends on the wave generation model used.

The formulation (3.12) is now checked against the deep water model given in the SPM (1984). In deep water (3.13) reduces to:

$$V_g = \frac{g}{4\pi} T_m$$

Then substituting in the SPM relationship between T_m and fetch, F :

$$V_g = \frac{U_A}{g} \frac{g}{4\pi} 0.286 \left(\frac{gF}{U_A^2} \right)^{1/3} \quad (3.14)$$

Substituting into (3.12) gives

$$t_{lim}(F) = \int_0^F \frac{4\pi}{U_A} \frac{1}{0.286} \left(\frac{g}{U_A^2} \right)^{-1/3} (F^*)^{1/3} dF^*$$

$$t_{lim}(F) = \frac{4\pi}{0.286 U_A} \frac{3}{2} \left(\frac{g}{U_A^2} \right)^{-1/3} F^{2/3}$$

And non-dimensionalising:

$$\frac{g}{U_A} t_{lim}(F) = 65.9 \left(\frac{gF}{U_A^2} \right)^{2/3} \quad (3.15)$$

This can be compared with the result given in the SPM (1984), i.e. eqn. (3.3):

$$\frac{g}{U_A} t_{lim}(F) = 68.8 \left(\frac{gF}{U_A^2} \right)^{2/3} \quad (3.16)$$

These results are within 5% of each other. Note that these equations are only valid up to the fully developed state as they are based on the relationship (3.14) between T_m and F , and this is not valid for fully developed seas.

The equation given in the SPM (1984) for t_{lim} for development in shallow water is not so good. Figure 3 shows a plot of the limiting storm duration against depth calculated using the SPM (1984) model for a fetch of 100 km and a windspeed of 15 m/s ($U_A = 22.3$ m/s). Also shown is a curve giving the minimum time it would take for the waves to cross the fetch based on the final group velocity, v_g :

$$t_{min} = t_{vg} = F/v_g \quad (3.17)$$

This curve should be considered to represent the minimum limiting duration since it is based on the group velocity of the waves at the end of the fetch. The wave period of the sea state for shorter fetch lengths would have been lower corresponding to a lower group velocity and therefore a longer duration. In deep water the shallow water t_{lim} agrees reasonably well with the deep water t_{lim} . As would be expected it is somewhat longer than t_{vg} . However for water depths below 10 m the calculated values of the limiting duration become unrealistically low. In a water depth of 1 m the limiting duration given by the shallow water formulation suggests that the speed of the sea state is ten times

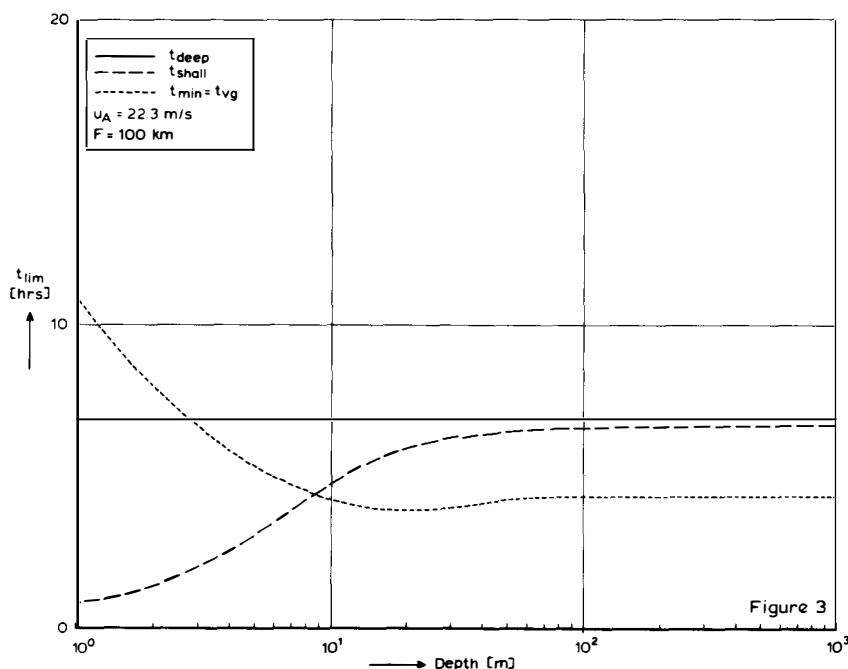


Fig. 3. Variation of limiting duration with depth predicted using various methods.

greater than the group velocity of the waves. Some revision of the formulation seems to be needed.

4. REVISED FORMULATIONS

4.1. Introduction

A single set of equations has been derived to replace the three sets given in the SPM (1984). The advantage of this reduced number of equations is two-fold. They are easier to apply (no decisions about whether the water is deep or whether the sea state is fully developed) and sudden changes in results in the transition between formulae are eliminated. Such step changes can be very inconvenient in some applications.

4.2 Theoretical approach

The starting point of the derivation of the new formulation was that the SPM model is correct in deep water, in shallow water, for small fetch lengths and for large fetch lengths.

A function with appropriate behaviour was constructed and the coefficients in the function were then adjusted to asymptotically match the SPM model for low and high fetch lengths and water depths. The formulation so obtained was tested to ensure that it agreed sufficiently closely with the SPM formulations in situations where the water is clearly shallow or clearly deep.

4.3 Revised equations for H_s and T_m (unlimited duration of time)

The modifications presented here have not been based on any theory of wave development or decay but have been formulated to asymptotically match the existing equations in the areas where there is most confidence in these. Some small (less than 3%) errors in asymptotic values reached have been allowed in some cases to get better agreement with formulations at intermediate values. The revised equations are:

$$\tilde{H}_{\text{rev}} = 0.25 \tanh(0.6\tilde{d}^{0.75}) \tanh^{0.5} \left[\frac{4.3 \times 10^{-5} \tilde{F}}{\tanh^2(0.6\tilde{d}^{0.75})} \right] \quad (4.1)$$

$$\tilde{T}_{\text{rev}} = 8.3 \tanh(0.76\tilde{d}^{0.375}) \tanh^{1/3} \left[\frac{4.1 \times 10^{-5} \tilde{F}}{\tanh^3(0.76\tilde{d}^{0.375})} \right] \quad (4.2)$$

The behaviour of these equations is first examined asymptotically and then over the whole range of values, using graphs.

The following two relationships are used constantly:

$$\tanh(x) \approx x \quad |x| \ll 1$$

$$\tanh(x) \approx 1 \quad x \gg 1$$

Then in deep water with low fetch ($\tilde{d} \gg 1, \tilde{F} \ll 10,000$) 4.1 and 4.2 become:

$$\tilde{H}_{\text{rev}} \approx 0.0016\tilde{F}^{0.5} \quad (4.3)$$

$$\tilde{T}_{\text{rev}} \approx 0.286\tilde{F}^{1/3} \quad (4.4)$$

These approximations are in complete agreement with those for the SPM (1984) deep water fetch limited case (see eqns. 3.1 and 3.2) and in fact the relationships can remain remarkably close for fetch lengths up to hundreds of kilometres. In deep water, for long fetch lengths, 4.1 and 4.2 become:

$$\tilde{H}_{\text{rev}} \approx 0.25 \quad (\tilde{d} \gg 1, \tilde{F} \gg 10,000) \quad (4.5)$$

$$\tilde{T}_{\text{rev}} \approx 8.3 \quad (\tilde{d} \gg 1, \tilde{F} \gg 10,000) \quad (4.6)$$

These values are 2%–3% greater than the deep water fully developed values given in the SPM (1984), see eqns. 3.6 and 3.7.

In shallow water with long fetch lengths ($\tilde{d} \ll 1, \tilde{F} \gg 1$), the revised formulations can be approximated by:

$$\tilde{H}_{\text{rev}} \approx 0.25 \times 0.6\tilde{d}^{0.75} = 0.15\tilde{d}^{0.75} \quad (4.7)$$

$$\tilde{T}_{\text{rev}} \approx 8.3 \times 0.76\tilde{d}^{0.375} = 6.3\tilde{d}^{0.375} \quad (4.8)$$

The SPM (1984) shallow water equations can be similarly approximated and eqns. (3.8) and (3.9) become:

$$\tilde{H}_s \approx 0.2835 \times 0.53\tilde{d}^{3/4} = 0.15\tilde{d}^{0.75} \quad (4.9)$$

$$\tilde{T}_m \approx 7.54 \times 0.833\tilde{d}^{3/8} = 6.3\tilde{d}^{0.375} \quad (4.10)$$

And thus there is again agreement to two significant figures.

Finally, on asymptotics, in shallow water, with very, very short fetch lengths, the water is effectively deep and the finite depth terms cancel in the shallow water equations and the three sets of equations agree (shallow water, deep water fetch limited, and revised).

In Fig. 1 a comparison is made between the variation of H_s with fetch in deep water predicted by the revised equations, the SPM (1984) deep water equations and the SPM (1984) shallow water equations (using a depth of 10,000 m). The windspeed was 15 m/s ($U_A = 22.3$ m/s).

There is good agreement of all three formulations at low fetch lengths. The shallow water formulation does not agree well with the deep water formulation for fetch lengths greater than about 500 km, whereas the revised formulation lies close to the deep water formulation except close to the transition between fetch limited and fully developed seas. Further, the revised formulation generally lies between the results of the shallow and deep water formulations.

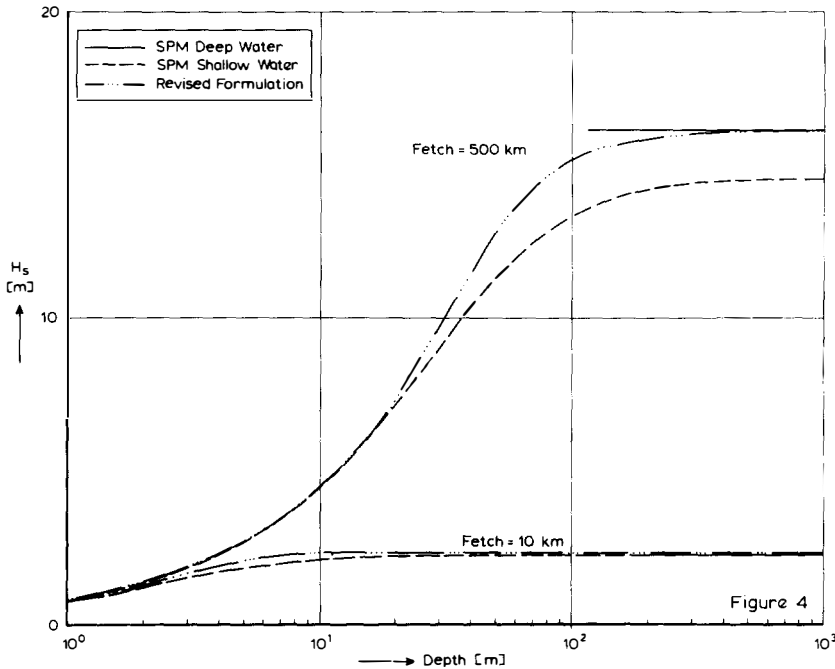


Fig. 4. Variation of predicted significant wave height with depth for various fetch lengths (unlimited duration, $U_A \approx 22.3$ m/s).

Figure 4 provides further support for the modified formulations. It shows the variation of H_s with depth, predicted by the three formulations for a fetch of 10 km and 500 km. In the 500 km case it agrees well with the shallow water formulation up to a depth of about 20 m and also matches well with the deep water solution. However, for a fetch of 10 km, while there is good agreement between results of the three formulations for high depths, the revised formulation gives somewhat higher results than the shallow water formulation at intermediate depths.

Figures 2 and 5 provide similar comparisons with similar results for the predictions made for T_m .

In conclusion, in most situations the revised formulations provide a good approximation to the relevant SPM (1984) formulation. At intermediate parameter values the revised formulation gives a smooth transition between the cases.

4.4 Possible revisions for the limiting duration

The most important consideration in this section is that the parameters H_s and T_m are only really sensitive to changes in the fetch for relatively low fetch

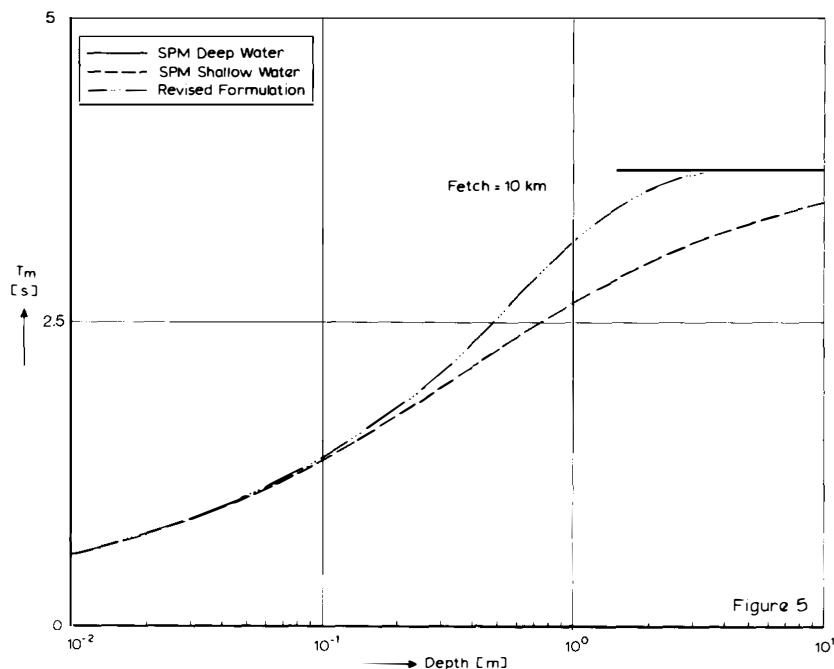


Fig. 5. Variation of predicted period of peak spectral energy with depth for a fetch of 10 km (unlimited duration, $U_A = 22.3$ m/s).

lengths. Further at low fetch lengths the wave length is not yet large enough compared with the water depth for significant shallow water effects on the group velocity. Thus the expression used for V_g in section 3.3 to get expression 3.15 can be used to approximately cover low water depths. The accuracy of this expression is now compared with a more exact expression which must be numerically integrated.

Given the assumption that wave energy can be represented by the group velocity associated with the period of peak spectral energy, the exact mathematical expression for the limiting duration is:

$$t_{\text{lim}}(F) = \int_0^F \frac{1}{V_g(F^*)} dF^* \quad (4.11)$$

$$\text{where: } V_g(F) = \left(\frac{1}{2} + \frac{kh}{\sinh 2kh} \right) \frac{\omega}{k}$$

$$\omega = 2\pi/T_{\text{rev}}$$

$$k \tanh kh = \omega^2/g$$

$$h = \text{water depth}$$

and T_{rev} is derived according to eqn. (4.2).

The results of numerically integrating expression 4.11 were compared with

the results of expression 3.15 for various fetch lengths, water depths and wind speeds. As expected, agreement was reasonably good at low fetch lengths but for larger fetch lengths differences of up to 15% were found in the predicted limiting duration. However when the effect of this error on the predicted significant wave height and period of peak spectral energy was tested, the agreement was always within 3% and usually much closer. This test was done by substituting the duration found in the numerical integration of 4.11 into the inverted form of equation 3.15. The resulting fetch was then used to calculate H_{rev} and T_{rev} according to equations 4.1 and 4.2 and compared with the results obtained with the original fetch.

The range of depth used was between 1 m and 100 m, fetch between 0.5 km and 10,000 km and wind stress factors of 10, 22.3 and 50 m/s were used. This presents a wide cross section of the conditions likely to be encountered and an error of 3% is reasonable for the level of sophistication of this model. It is therefore recommended that the expression 3.15 should be used for the limiting duration in all circumstances:

$$\tilde{t}_{rev} = 65.9\tilde{F}^{2/3} \quad (4.12)$$

resulting in a modified fetch length

$$\tilde{F}' = \left(\frac{\tilde{t}}{65.9} \right)^{3/2} \quad \text{for } \tilde{t} < \tilde{t}_{rev} \quad (4.13)$$

Note that this gives slightly conservative results by comparison with original SPM (1984) formulations as it gives limiting durations about 5% shorter than those predicted in the SPM (1984) deep water model. Comparison was not made with the SPM (1984) shallow water model.

Other more complicated and more accurate approximations can be derived which account for the variation of group velocity with water depth. However the drawback of such expressions is that they cannot be easily inverted, whereas this is simple for expression 4.12.

5. CONCLUSIONS

The SPM (1984) model relies on several sets of formulations, the set to be used depending on the input parameters. There are two deep water formulations, for fetch limited seas and for fully developed seas. These do not match at the transition point given in the formulations. Furthermore at large fetch lengths the shallow water formulations do not match the deep water formulations with increasing water depth.

An alternative single set of equations for significant wave height and period of peak spectral energy has been derived, which closely asymptotically match the SPM (1984) formulations and provide a smooth transition between them. The advantage of these formulations is that they are easier to apply and probably give more reliable results in the transition between the various sets of equations in the SPM (1984) model.

A slightly modified (more conservative) version of the expression for the

limiting duration in deep water is given for situations where the sea state may be limited by the duration of a storm rather than the fetch. It is recommended that this expression should be applied for both deep and shallow water as it is both easier to apply and gives more reliable results.

REFERENCE

U.S. Army, 1984. Shore Protection Manual, Vol. 1. Coastal Engineering Research Center.